# SRA-VFM: Boosting Remote Sensing Change Detection via Slice-Reassembled Augmentation and Vision Foundation Model-Guided Dual Streams

Wenhui Ye, Weimin Lei, *Member, IEEE*, Wenchao Zhang, and Wei Zhang

***Abstract*—To address the dual challenges of inadequate deep semantic feature representation and limited-data diversity in bi-temporal remote sensing change detection (RSCD), we propose a collaborative optimization framework [SRA-vision foundation model (VFM)] integrating customized data augmentation and hierarchical feature interpretation. SRA-VFM comprises three core components: slice-reassemble augmentation (SRA), a dual-stream feature encoding–decoding (DFED) network, and a multitask segmentation head. The SRA module synthesizes diverse training samples through random slicing and semantic reassembly while preserving local feature consistency. The dual-stream encoder, based on the FastSAM pretrained model, incorporates a top-down feature adapter to align pretrained features with remote sensing data distributions via cross-level fusion and low-dimensional semantic mapping. The bottom-up decoder leverages semantically aligned pyramid features for progressive upsampling, restoring high-resolution spatial details, and enhancing multiscale representation. The multitask head jointly optimizes change region detection and edge refinement, accelerating convergence and improving boundary localization. Extensive experiments on five benchmark datasets (LEVIR-CD, WHU-CD, CLCD, S2Looking, and SYSU-CD) demonstrate SRA-VFM's superiority over state-of-the-art (SOTA) methods, achieving m$F1$/mIoU of 96.01%/92.55%, 97.13%/94.53%, 88.91%/81.21%, 83.13%/48.91%, and 88.36%/79.67%, respectively. The code will be publicly available upon publication.**

***Index Terms*—Change detection (CD), data augmentation, remote sensing image (RSI), vision foundation models (VFMs).**

## I. Introduction

REMOTE sensing change detection (RSCD) aims to identify and quantify changes in urban construction, surface coverage, or land use by analyzing remote sensing images (RSIs) of the same area taken at different time points. This technique is a critical component of geographic information science and has played a pivotal role in various fields, including environmental monitoring [1], urban planning [2], disaster response [3], [4], and resource management [5], [6].

RSI surpasses natural scenes in both spatial extent and background complexity, embedding multiscale content from regional landforms to object-level details. Consequently, RSCD must jointly capture macroscopic surface transitions and fine-grained alterations, necessitating scale-aware feature extraction and matching. The dual-input, bi-temporal paradigm further intensifies the task; it demands explicit modeling of spatiotemporal correspondence, a requirement that current methods rarely fulfill, thereby constraining detection accuracy and robustness.

Simultaneously, RSI acquisition is expensive and conditioned on stringent logistics; the bi-temporal requirement of RSCD further aggravates sample scarcity, severely limiting the efficacy of data-hungry deep models. Data augmentation mitigates this bottleneck by synthetically expanding the training set while preserving label fidelity. Conventional strategies encompass geometric (rotation, translation, and shear) and nongeometric (noise, flip, crop, and color jitter) perturbations (Cutout [7], RandomErasing [8], and Hide-and-Seek [9]); yet random erasures risk contextual loss, an issue partially resolved by structured masks such as GridMask [10]. These techniques inflate sample counts but fail to enrich semantic diversity. Consequently, feature-level mixing approaches have emerged: SalfMix [11] leverages saliency-guided intraimage recombination, whereas intersample methods (Mixup [12] and its extensions [13], [14], [15], [16]) generate virtual examples via linear interpolation of inputs and labels, albeit at the cost of spatial interpretability. Automated augmentation has also been framed as a discrete search problem [17], [18]. A remaining challenge for RSCD is to devise augmentation schemes that simultaneously diversify appearances and maintain coherent bi-temporal change patterns.

Although the above methods perform well on natural images, given the characteristics of RSIs, such as complex backgrounds, multiple interferences, diverse target classes, and small pixel proportions, conventional data augmentation methods may not be effective, necessitating tailored approaches for their specific features. The ITSA [19] method proposes a new iterative training sample augmentation strategy in the field of land cover change detection (CD), which iteratively explores potential training samples near initial samples. IAug_CDNet

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[20] uses a conditional generative adversarial network to generate building instances based on input semantic layouts. Especially for RSCD tasks, they not only require extracting foreground regions from bi-temporal input images but also necessitate the network's ability to identify "changes." Unlike existing image data augmentation methods, we emphasize that for RSCD tasks, it is not only necessary to enhance the semantic richness of the data but also to expand the "change" diversity of the dataset. Therefore, in this article, we propose the slice-reassembly augmentation (SRA) data augmentation method for RSCD tasks, aiming to enable the network to fully learn how to identify change features.

Recent advances in vision foundation models (VFMs) offer new opportunities for RSCD. Researchers have opened up a new pathway through in-depth exploration of VFM, aiming to enhance the basic feature extraction capabilities of neural networks. VFM is capable of mining potential knowledge from vast general datasets. However, when applying them to the field of RSIs with complex spatiotemporal backgrounds, further exploration and research are still required. Models like SpectralGPT [21], RSBuilding [22], and Change-Agent [23] demonstrate the potential of pretrained knowledge transfer. FoMo [24] proposes the first unified forest monitoring benchmark (FoMo-Bench) and a pretraining framework (FoMo-Net) supporting arbitrary remote sensing modality combinations, significantly enhancing large-scale models' generalization and adaptation in global multiscale/multimodal forest monitoring tasks. Prithvi [25], the first large-scale geospatial foundation model, realizes rapid generalization and precise application to multiresolution, multitemporal global remote sensing tasks under scarce-label conditions through self-supervised pretraining and efficient data sampling.

Similarly, numerous scholars investigate the application of VFMs in RSCD tasks. TTP [26] introduces low-rank trainable parameters to mitigate spatial semantic shifts resulting from the segment anything model (SAM) [27] model in the RSIs domain while presenting a time-travel activation gate to enable the model to focus on capturing effective changes between bi-temporal images. BAN [28] extracts general features using a frozen foundational model and designs a bridging module to selectively resample general knowledge, injecting it into designed bi-temporal adapter branches for knowledge transfer. SAM-CD [29] employs a frozen FastSAM for feature extraction and introduces an additional semantic learning branch, implicitly supervising the learning of semantic latent representations in bi-temporal RSIs through temporal constraints. ChangeCLIP [30] reconstructs the original contrastive language-image pretraining (CLIP) model to extract bi-temporal features and proposes a novel differential features compensation module to capture detailed semantic changes between images across two time points. Remoteclip [31], based on the CLIP strategy, aligns the representations of large-scale image–text pairs through contrastive learning. Meta-CD [32], which freezes the FastSAM backbone, integrates a lightweight convolutional neural network (CNN) adapter and introduces a pixel-wise binarization module. The framework delivers robust CD using only 5%–30% of the annotated data, markedly reduces data dependence, and enhances both accuracy and zero-shot transferability in limited-data scenarios.

In summary, current VFM-based RSCD methods generally utilize VFM as a fixed-parameter encoder to achieve more powerful feature extraction efficiency. However, when facing RSCD tasks, a notable issue arises: a significant semantic gap exists between the low-level semantic information extracted by VFM encoders trained on natural images and the high-level semantic information required by decoder ends. In light of this, we propose strategies for cross-level feature fusion and low-dimensional semantic space mapping to effectively address the differences between pretrained features and the data distribution in the RS domain. Additionally, this study introduces multilevel feature interaction and a multitask segmentation head method, aiming to simultaneously improve the detection accuracy of changed areas and the robustness of boundary localization.

Addressing the aforementioned issues, we propose a novel data-model collaborative optimization approach, termed SRA-VFM, to enhance the performance of RSCD tasks. Specifically, we tailor an SRA method for the RSCD task to enhance the network's generalization capability and its ability to discern semantic changes. Furthermore, we construct a dual-stream feature encoding–decoding (DFED) network and a multitask-driven segmentation head (MSH) to optimize the network structure, thereby bridging the semantic gap between the VFM and RSCD tasks and accelerating network convergence.

Overall, our main contributions are as follows.

1) A slice-reassemble strategy is proposed to enhance data diversity while preserving critical feature correlations via image reassembly constrained by local semantic consistency, boosting model generalization capability.
2) A pretrained model-based dual-stream encoder and feature adaptation module are designed to bridge the gap between general visual features and remote sensing data distributions, enabling the extraction of highly discriminative features.
3) Our framework integrates a pyramid decoder with a dual-task segmentation head, enhancing detection precision of altered regions and boundary localization stability through cross-scale contextual fusion and refined edge constraints.

The rest of this article is structured as follows. In Section II, we detail the method we propose. In Section III, we experimentally validate the effectiveness of our method and present visual results. Finally, in Section IV, we summarize the entire article.

## II. Our Method

In this section, we will provide a detailed description of the proposed network structure. First, we will analyze the overall architecture of the network. Subsequently, we will elaborate on the three components proposed in this article: SRA, DFED, and MSH. Finally, we will present the loss functions employed for network training.

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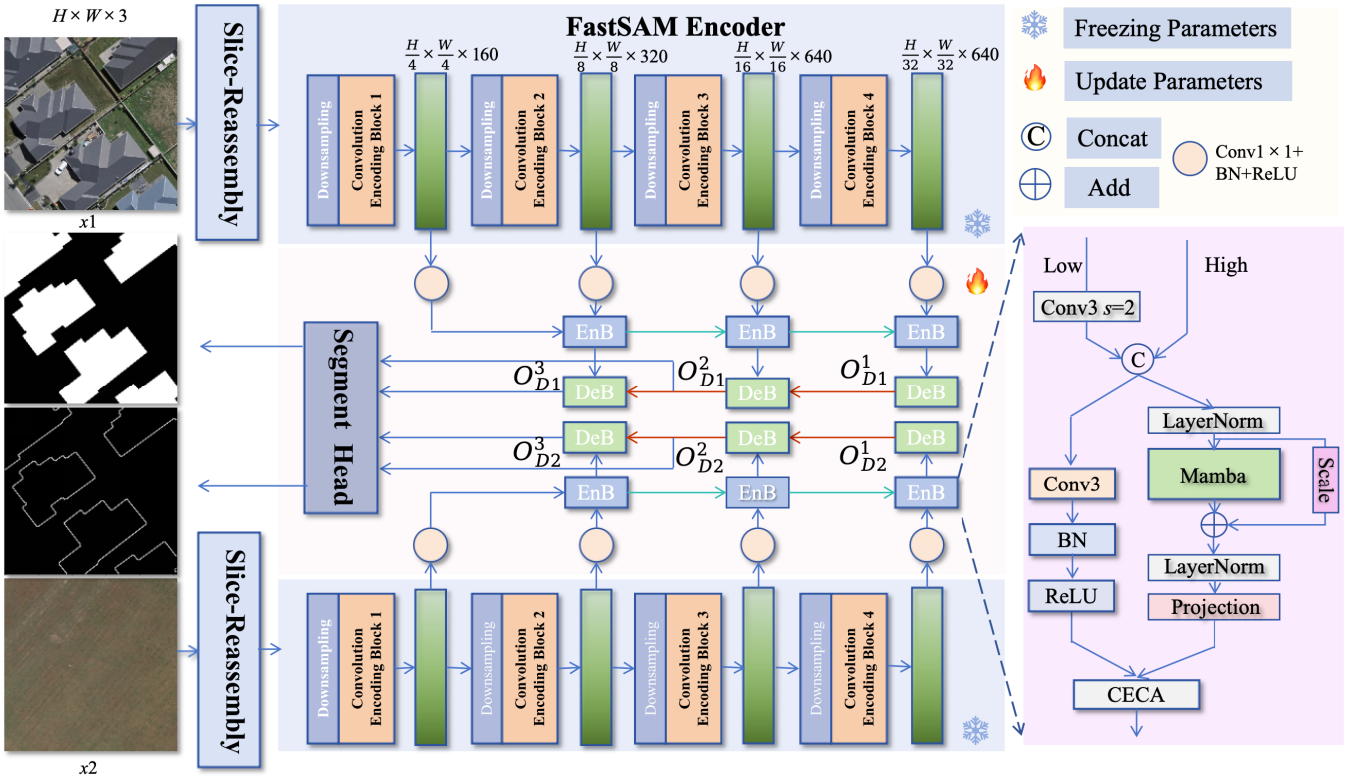



Fig. 1. SRA-VFM network architecture. It is composed of slice-reassembly augmentation, DFED, and multitask segmentation head. It is worth noting that the parameters of dual-stream VFM encoder are shared.

### A. Overall Architecture

As shown in the Fig. 1, to obtain more discriminative deep semantic features from bi-temporal RSIs, we adopted a dual-stream architecture for CD. Our network consists of four main components: VFM (FastSAM backbone) static encoder, top-down feature adapter, bottom-up feature decoder, and change region segmentation head. The VFM (FastSAM backbone) static encoder extracts basic features from RSIs using its pretrained weights on large-scale datasets, facilitating change feature interpretation. The top-down feature adapter fuses semantic features from different layers to fine-tune general base features, mapping them to a common low-dimensional feature space for semantic alignment. This balances model generality and task specificity, making the extracted features better suited to the distribution of RSI data. The bottom-up feature decoder upsamples aligned pyramid features to restore high-dimensional spatial details. It fuses low-level features to enhance semantic context interaction and improves the network's ability to extract multiscale features through multilayer pyramid feature flow. The change region segmentation head performs multitask learning and feature interaction enhancement by combining CD and edge-assisted CD, promoting feature migration and aiding network convergence.

Given the dual-stream symmetric architecture adopted in our design, the static encoder, top-down feature adaptor, and bottom-up feature decoder within the VFM (FastSAM backbone network) are parameter-shared across both branches. We will therefore illustrate the feature extraction workflow during CD using the $X1$ image processing stream as a representative case study. First, we input the bi-temporal RSIs pair $X1, X2 \in \mathbb{R}^{H\times W\times 3}$ from the same region at different times into the VFM (FastSAM) static encoder to extract features at multiple scales $V_i \in \mathbb{R}^{(H/s_i)\times(W/s_i)\times C_i}$, where $i \in 1, 2, 3, 4$ represents different feature stages, $s_i \in 4, 8, 16, 32$ represents the downsampling ratio for each stage, and $C_i \in 160, 320, 640, 640$ represents the number of channels in each feature map. Note that these network parameters are directly loaded with pretrained FastSAM weights and remain fixed during training, without gradient backpropagation. Next, the top-down feature adapter receives the multilevel features $V_i$ and performs further feature mapping and alignment to produce features $E_i \in \mathbb{R}^{(H/s_i)\times(W/s_i)\times(C_i/r)}$, where $r$ represents the channel compression ratio during feature mapping. Then, the bottom-up feature decoder fuses low-level features and uses deconvolutional upsampling to generate output features $D_1 \in \mathbb{R}^{(H/s_1)\times(W/s_1)\times C_1}$. Similarly, we obtain the output features $D_2$ for the $X2$ image. Finally, we send the extracted features $D_1$ and $D_2$ from the two symmetric branches to the change region segmentation head network for deep fusion and discrimination, resulting in the CD prediction mask $M \in \mathbb{R}^{H\times W\times 1}$ and the change region boundary mask $B \in \mathbb{R}^{H\times W\times 1}$.

### B. Slice-Reassembly Augmentation

To increase the diversity of training samples and improve the network's generalization capability, we designed an SRA module to preprocess the input images and corresponding change mask labels before feeding them into the network model. Specifically, as shown in the Fig. 2, we first uniformly

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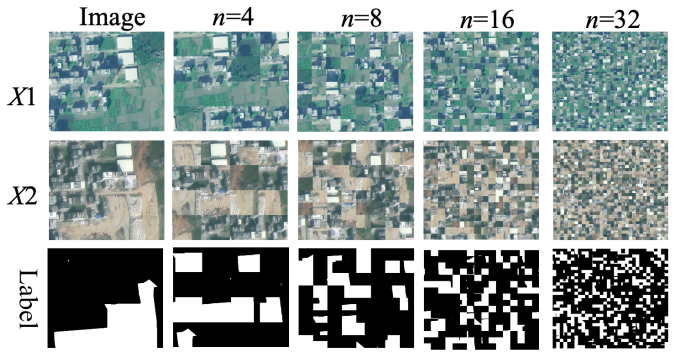



Fig. 2. Architecture of the proposed SRA, $n$ represents different numbers of slices.

divide the input images $X1$ and $X2$ into $n \times n$ sized image patches. We then perform patch-wise slicing using random sampling and reassemble these patches into a new RSI. Additionally, we set a random seed to ensure that $X1$, $X2$, and the corresponding change mask labels use the same slicing order during each processing step, maintaining consistency between training image pairs and their labels. It should be emphasized that an identical seed is used for both images and the mask to preserve exact alignment; different seeds would produce inconsistent patch–label pairs and cause training failure.

Unlike traditional data augmentation methods such as random flipping, random cropping, and hue adjustment, the SRA module we designed is a specialized and efficient data augmentation technique tailored for RSCD. The main idea is to increase data richness through reassembly, enabling the network to learn how to identify change features effectively. Although this method disrupts some domain-specific dependencies in the original images, it maintains semantic relevance in local regions. We believe that the network's ability to recognize changes and its capacity to learn semantic relevance are equally important for CD tasks. This belief forms the basis of our data augmentation method design. Finally, we validated the effectiveness of this method through experiments. We also explored the impact of different patch sizes on network performance, which will be discussed in Section III.

The proposed slicing-reassembly data augmentation method employs synchronous slicing and reassembly of bi-temporal RSIs and corresponding labels. While disrupting global spatial continuity, this approach enforces the establishment of local spatiotemporal correspondences, preserving semantic coherence in changed regions while simulating discontinuous spatiotemporal variations of real-world land cover distributions. By deliberately decoupling the local spatial dependencies relied upon by traditional CNNs, the strategy compels the model to develop cross-spatial temporal feature association capabilities and mechanisms for learning discontinuous regional change patterns. This data-level intervention directly enhances the CD network's capacity to model complex spatiotemporal relationships, creating a training environment more aligned with practical scenarios such as cross-region or cross-sensor applications.

### C. Dual-Stream Feature Encoding–Decoding

In order to process bi-temporal RSIs simultaneously, we have adhered to a symmetrical dual-stream design paradigm and constructed the DFED module for feature encoding and decoding. As mentioned earlier, we used FastSAM to extract features from the encoder. In the following, we will focus on explaining the decoder. To make the features extracted by the VFM static encoder suitable for RSI tasks and reduce the semantic gap with the decoder, we adopted the feature pyramid network (FPN) design paradigm and used a top-down fusion approach to combine different levels of $V_i$. First, to reduce computational resource usage and further condense the features, we performed feature channel compression on each $V_i$ to obtain more compact multilevel features $Z_i$. The process can be described by the following equation:

$$Z_i = \delta\left(\mathrm{BN}\left(F1_r^k(V_i)\right)\right) \tag{1}$$

where $F1_r^k(\cdot)$ represents a convolutional layer with a kernel size of $k$ and a channel compression rate of $r$. In this convolution, we set $k = 1$ and $r = 2$. $\mathrm{BN}(\cdot)$ represents the batch normalization (BN) layer and $\delta$ represents the ReLU activation function.

Next, we input $Z_i$ into the EnB module for feature fusion. Each encoding block (EnB) module processes two different level features: a lower level feature $Z_L$ with a larger scale and a higher level feature $Z_H$ with a smaller scale. To obtain the same feature scale, we use a convolutional layer to compress the scale of $Z_L$ to $P_L$. The process can be described by the following equation:

$$P_L = \delta\left(\mathrm{BN}\left(F2_s^k(Z_L)\right)\right) \tag{2}$$

where $F2_s^k(\cdot)$ represents a convolutional layer with a kernel size of $k$ and a stride of $s$. In this convolution, we set $k = 3$ and $s = 2$.

Next, we merge the features $P_L$ and $Z_H$ of the same size along the channel dimension. The merged feature then passes through a dual-channel feature extractor composed of parallel convolutions and residual Mamba blocks to enhance detail features and long-distance global features separately, resulting in $\mathrm{PV}_L$ and $\mathrm{GZ}_H$. The process can be described by the following equations:

$$M = \mathrm{Concat}\left(P_L, Z_H\right) \tag{3}$$

$$\mathrm{DP}_L = \delta\left(\mathrm{BN}\left(F3_s^k(M)\right)\right) \tag{4}$$

$$\mathrm{GZ}_H = \mathrm{Proj}\left(\mathrm{LN}\left(\mathrm{Mamba}\left(\mathrm{LN}\left(M\right)\right) + \mathrm{scale} \times M\right)\right) \tag{5}$$

in the convolution $F3_s^k(\cdot)$, we set $k = 3$ and $r = 1$. $\mathrm{LN}(\cdot)$ represents the layer normalization module, and scale represents the residual scaling coefficient, which we set to 1. Mamba represents the state space sequence layer, and $\mathrm{Proj}(\cdot)$ represents the fully connected projection layer.

Finally, we fuse the detail features $\mathrm{DP}_L$ and global features $\mathrm{GZ}_H$ using the interactive attention mechanism ECA [33]

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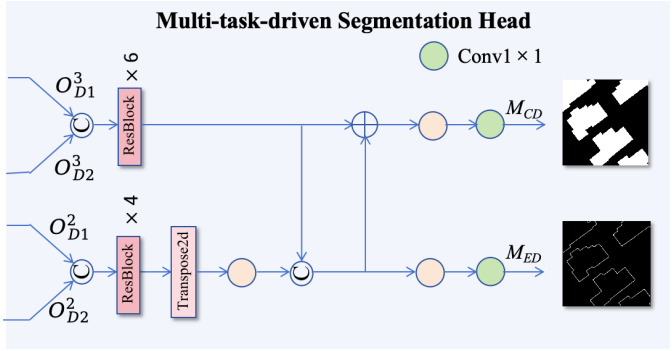



Fig. 3. Architecture of the proposed multitask segmentation head. In addition to the inherent CD task, we also added the edge-assisted segmentation task to promote the network's attention to the contour of the changed region.

module to obtain the output feature $O_E$. The process can be described by the following equation:

$$O_E = \sigma\left(F1d\left(\text{AvgP}\left(\text{DP}_L\right)\right)\right) \times \text{GZ}_H. \tag{6}$$

### D. Multitask Segmentation Head

After passing through the VFM (FastSAM backbone) static encoder, top-down feature adapter, and bottom-up feature decoder, $X1$ and $X2$ produce features $O^j_{D1}$ and $O^j_{D2}(j \in 1, 2, 3)$. In the MSH network, there are two types of learning tasks. First, for the CD task, we need to model pixel-level change region features. Second, to improve the completeness of object boundaries in the output change map, we designed an auxiliary task for detecting change region contours. For these two tasks, we extracted features from different stages of the decoder to implement multitask learning.

Specifically, as illustrated in Fig. 3, in the change region segmentation task, we use semantically enriched features $O^3_{D1}$ and $O^3_{D2}$. Following feature fusion, we apply six consecutive residual modules to generate deep change features, yielding feature CD. In the auxiliary task of change region contour edge detection, we leverage detail-preserving features $O^2_{D1}$ and $O^2_{D2}$. Similarly, channel-wise fusion of bi-temporal decoded features precedes four residual modules that model contour features, yielding feature ED. Subsequently, a cross-fusion mechanism facilitates gradient flow between tasks. Finally, the framework outputs predicted change region mask $M_{\text{CD}}$ and the change region contour edge mask $M_{\text{ED}}$. The process can be described by the following equations:

$$\text{MC}_{\text{ED}} = \text{Concat}\left(F4^k_{\text{BR}}\left(\text{ED}\right), \text{CD}\right) \tag{7}$$

$$M_{\text{ED}} = F6^k\left(F5^k_{\text{BR}}\left(\text{MC}_{\text{ED}}\right)\right) \tag{8}$$

$$M_{\text{CD}} = F8^k\left(F7^k_{\text{BR}}\left(\text{MC}_{\text{ED}} + \text{CD}\right)\right) \tag{9}$$

where $Fh^k_{\text{BR}}(\cdot), h \in \{4, 5, 7\}$ represents a convolution operation with a BN layer and rectified linear unit (ReLU) activation function; $F^k(\cdot)$ represents only the convolution layer; and $k$ represents the kernel size, which we set to 1.

### E. Loss Function

For both RSCD and change region contour edge detection tasks, the final predictions are binary masks. Therefore, during the training of the SRA-VFM network, we used the binary cross-entropy loss function to measure the distance between the predicted probabilities and true labels, and this was used as the optimization objective for the network model. Specifically, the overall loss function during training consists of two parts, as shown in the following:

$$L_{\text{all}} = L_{\text{BCE}}\left(M_{\text{CD}}, \text{GT}\right) + \beta \times L_{\text{BCE}}\left(M_{\text{ED}}, \text{EGT}\right) \tag{10}$$

where $L_{\text{BCE}}(\cdot, \cdot)$ represents the binary cross-entropy loss function, ground truth (GT) represents the change region labels corresponding to the input bi-temporal RSIs, and EGT represents the edge contour labels of the change region (obtained from GT). $\beta$ is a balancing factor for the two tasks, which we set to 10 in our experiments.

## III. Experiments

This section first outlines the experimental datasets, evaluation metrics, and hyperparameter settings utilized within the proposed framework. Following this, a comprehensive performance comparison is conducted against SOTA methods using five established benchmarks: LEVIR-CD, WHU-CD, CLCD, S2Looking, and SYSU-CD. Finally, structured ablation studies are performed to quantitatively verify the effectiveness of individual architectural innovations introduced in this work.

### A. Dataset and Metrics

*LEVIR-CD [34]:* LEVIR-CD is a widely utilized dataset for CD in RSIs, provided by the Learning, Vision and Remote Sensing Laboratory. It consists of 637 very high-resolution (VHR, 0.5 m/pixel) Google Earth (GE) bi-temporal image patch pairs covering a span of 5 to 15 years, each with dimensions of 1024 × 1024 pixels. The dataset has been annotated for building changes by remote sensing experts. The final dataset comprises 31 333 individual examples of building changes. To aid in the training and validation of model performance, these 637 image pairs are divided into three sets: 445 as the training set, 64 as the validation set, and 128 as the test set.

*WHU-CD [35]:* The WHU-CD dataset, provided by the SIGMA research team at Wuhan University, is a high-quality resource specifically designed for RSCD. It comprises two aerial image pairs, each measuring 32 507 × 15.354 pixels, captured in Christchurch, New Zealand, between 2012 and 2016. These images cover an urban area of approximately 20.5 square kilometers with a spatial resolution as fine as 0.2 m. To optimize computational efficiency, the images were cropped into patches of size 256 × 256, resulting in a total of 7620 bi-temporal image pairs. Following the methodology outlined in SAM-CD, we employed 6096 images for training, 762 images for validation, and 762 images for testing to evaluate the model's performance.

*CLCD [36]:* The CLCD dataset is a meticulously curated collection comprising 600 pairs of images capturing farmland changes. These images were acquired by the Gaofen-2 satellite over Guangdong Province, China, in 2017 and 2019. The dataset features high-quality images with spatial resolutions ranging from 0.5 to 2 m. Each pair consists of two 512 × 512

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pixel images along with corresponding binary labels indicating farmland changes. To facilitate efficient evaluation of model performance, the CLCD dataset is divided into three subsets: 320 pairs are designated for training, 120 pairs for validation, and the remaining 120 pairs for testing. Key land use/cover change types, including buildings, roads, lakes, and bare land, are annotated within these samples.

*S2Looking [37]:* The S2Looking dataset is a building CD dataset that encompasses large-scale side-view satellite images captured at various off-zenith angles. It consists of 5000 pairs of ultrahigh-resolution images acquired from the Gaofen, Hyper-Vision, and Beijing II satellites, each with dimensions of 1024 × 1024 pixels, spanning from 2017 to 2020. The dataset is divided into three subsets: 3500 pairs for training, 500 pairs for validation, and the remaining 1000 pairs for testing. The annotations in this dataset include 65 920 samples of newly constructed or demolished building areas across a variety of global scenarios, such as villages, farms, villas, retail centers, and industrial areas. The inclusion of large-scale side-view imaging and diverse scenes presents significant challenges for the task of building CD.

*SYSU-CD [38]:* The SYSU-CD dataset, released in 2022, is a comprehensive CD dataset comprising 20 000 pairs of RSIs of the Hong Kong region. These images have a spatial resolution of 0.5 m and span from 2007 to 2014. Each image pair measures 256 × 256 pixels. The dataset is divided into three subsets: 12 000 pairs are used for training, 4000 pairs for validation, and the remaining 4000 pairs for testing. Given that Hong Kong is a densely populated area with numerous high-rise buildings and is situated near the coastline, the scenarios included in this dataset capture both urban high-density constructions and port-related developments. This balance introduces a diverse set of change samples that are particularly relevant to real-world CD applications.

To comprehensively evaluate the performance of the algorithms, we employ five distinct evaluation metrics: precision (Pre), recall (Rec), overall accuracy (OA), mean $F1$ score (m$F1$), and mean intersection over union (mIoU). These metrics are derived from the interrelationships among true positives (TPs), false positives (FPs), true negatives (TNs), and false negatives (FNs). The specific formulations for these evaluation metrics are provided in the following equations:

$$sPre = \frac{TP}{TP + FP}, \quad Rec = \frac{TP}{TP + FN} \tag{11}$$

$$F_1 = 2 \times \frac{P \times R}{P + R}, \quad OA = \frac{TP + TN}{TP + FP + TN + FN} \tag{12}$$

$$IoU = \frac{TP}{TP + FP + FN}. \tag{13}$$

## B. Implementation Details

This study constructs an end-to-end network architecture based on the PyTorch deep learning framework, leveraging the computational resources of a single NVIDIA GeForce RTX 4090 Graphics Processing Unit for supervised training. During the data preprocessing phase, a three-tier augmentation strategy is designed by integrating random flipping, dynamic cropping, and the proposed SRA method, enhancing sample diversity in a data-driven manner. Differentiated cropping strategies are formulated according to the characteristics of various optical remote sensing datasets: For LEVIR-CD and S2Looking datasets (1024 × 1024 resolution), adaptive window cropping with 512 × 512 dimensions is applied; CLCD dataset (512 × 512 resolution) undergoes refined patch partitioning using 256 × 256 segments; WHU-CD and SYSU-CD datasets (256 × 256 resolution) retain native resolution inputs. Experimental configurations strictly adhere to standardized protocols, with all models trained through 50 epochs using a stochastic gradient descent (SGD) optimizer. The initial learning rate is set to 0.2, maintaining hyperparameter consistency across experiments to systematically evaluate the generalization capability of the proposed algorithm.

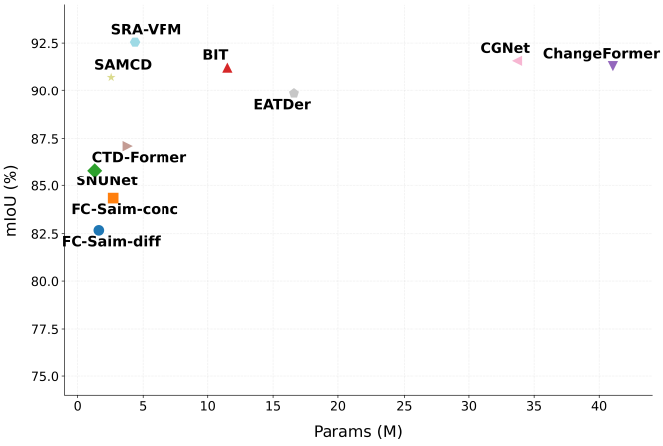



Fig. 4. Parameters-mIoU (Params-mIoU) performance comparison of evaluated methods on the LEVIR-CD dataset.

## C. Main Results

To comprehensively validate the effectiveness of our proposed algorithm, we conducted a comprehensive comparison analysis. As evidenced by the comparative analysis in Fig. 4, our method achieves SOTA performance. Specifically, we compared our method with several SOTA algorithms on five challenging datasets. These SOTA algorithms can be categorized into the following groups: CNN-based CD methods such as FC-Siam-Diff, FC-Siam-Conv [39], and SNUNet [40]; and Transformer-based methods like ChangeFormer [41], CTD-Former [42], and EATDer [43]; and VFM-based methods such as SAM-CD [29]. Additionally, to explore the balance between accuracy and computational complexity in our method, we selected two lightweight RSCD methods, light-CD [44] and tiny-CD [45], for comparison. This step aims to evaluate whether our method achieves high accuracy with low computational cost, thereby supporting its practical applicability.

*1) Model Complexity Analysis:* First, we compared the complexity of different algorithms from multiple dimensions in Table I, including parameter count, giga floating-point operations per second (GFLOPs), frames per second (FPS), model size, and mIoU on the LEVIR-CD dataset. The comparison results show that our method performs well in terms of overall performance. Taking SAM-CD as the baseline, SRA-VFM

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TABLE I

COMPARISON OF MODEL COMPLEXITY (NOTE: PARAMS (M) DENOTES THE NUMBER OF LEARNABLE PARAMETERS; BOLD VALUES INDICATE OPTIMAL OUTCOMES, AND UNDERLINED VALUES REPRESENT THE SECOND-BEST PERFORMANCES)

| Method | Params(M) | GFLOPs | FPS(s) | Model_Size(M) | mIoU |
|---|---|---|---|---|---|
| FC-Siam-diff | 1.66 | 21.45 | **1504.16** | 20.3 | 82.67 |
| FC-Saim-conc | 2.74 | 8.77 | 1312.02 | 23.1 | 84.34 |
| SNUNet | 1.35 | 18.91 | 156.74 | 148.2 | 85.84 |
| BIT[46] | 11.47 | 19.60 | 765.22 | 38.8 | 91.23 |
| ChangeFormer | 41.03 | 811.15 | 142.92 | 170.7 | 91.29 |
| CTD-Former | 3.85 | 104.62 | 445.34 | 53.9 | 87.11 |
| CGNet | 33.68 | 82.23 | 196.53 | 148.9 | 91.58 |
| EATDer | 16.59 | 23.43 | 133.65 | 28.7 | 89.84 |
| SAM-CD | 2.59 | 8.59 | 217.65 | 10.4 | 90.71 |
| LightCD-S | 0.35 | 1.65 | 705.68 | 8.5 | 90.49 |
| TinyCD | **0.28** | **1.54** | 752.14 | **7.2** | 90.75 |
| SRA-VFM | 4.39 | 9.87 | 210.16 | 67.7 | **92.55** |

TABLE II

RESULTS ON THE LEVIR-CD DATASET

| Method | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| FC-Siam-diff | 92.93 | 87.02 | 98.15 | 89.73 | 82.67 |
| FC-Siam-conc | 92.03 | 89.82 | 98.28 | 90.89 | 84.34 |
| SNUNet | 93.83 | 90.11 | 98.50 | 91.88 | 85.84 |
| BIT | 90.27 | 83.37 | 97.60 | 86.46 | 91.23 |
| ChangeFormer | 91.46 | 86.31 | 97.95 | 88.69 | 91.29 |
| CTD-Former | 93.60 | 91.85 | 98.62 | 92.71 | 87.11 |
| CGNet | 95.95 | 94.95 | 99.13 | 95.44 | 91.58 |
| EATDer | 96.29 | 92.70 | 98.87 | 94.41 | 89.84 |
| SAM-CD | 96.39 | 93.57 | 99.05 | 94.93 | 90.71 |
| LightCD-S | 95.12 | 94.48 | 99.00 | 94.80 | 90.49 |
| TinyCD | 96.09 | 93.88 | 99.05 | 94.96 | 90.75 |
| ChangeCLIP | 93.40 | 90.67 | 99.20 | 95.79 | 92.18 |
| MambaBCD[47] | 89.24 | 87.57 | 99.06 | - | - |
| FTA-Net[48] | 91.79 | 92.68 | - | - | - |
| IAE-CDNet[49] | 93.67 | 92.58 | - | - | - |
| SRA-VFM | **96.54** | **95.49** | **99.24** | **96.01** | **92.55** |

TABLE III

RESULTS ON THE WHU-CD DATASET

| Method | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| FC-Siam-diff | 94.52 | 89.32 | 98.70 | 91.74 | 85.65 |
| FC-Siam-conc | 93.94 | 92.83 | 98.90 | 93.37 | 88.19 |
| SNUNet | 92.61 | 83.26 | 98.10 | 87.30 | 79.40 |
| BIT | 87.92 | 93.41 | 98.29 | 90.46 | 83.72 |
| ChangeFormer | 96.28 | 92.95 | 99.12 | 94.55 | 90.08 |
| CTD-Former | 96.74 | **97.03** | 99.50 | 96.89 | 94.11 |
| CGNet | **98.04** | 96.14 | **99.52** | 97.07 | 94.44 |
| EATDer | 95.57 | 93.06 | 99.01 | 94.28 | 89.64 |
| SAM-CD | 97.47 | 94.09 | 99.31 | 95.71 | 92.04 |
| LightCD-S | 66.17 | 75.03 | 93.45 | 69.43 | 60.07 |
| TinyCD | 71.44 | 68.33 | 95.29 | 69.74 | 60.87 |
| FTA-Net | 96.02 | 94.43 | - | - | - |
| SRA-VFM | 97.80 | 96.46 | **99.52** | **97.13** | **94.53** |

TABLE IV

RESULTS ON THE CLCD DATASET

| Method | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| FC-Siam-diff | 83.13 | 74.60 | 94.73 | 78.10 | 68.19 |
| FC-Siam-conc | 81.65 | 79.94 | 94.84 | 80.77 | 70.98 |
| SNUNet | 85.92 | 82.62 | 95.84 | 84.19 | 75.11 |
| BIT | 81.42 | 70.27 | 94.17 | 74.39 | 64.51 |
| ChangeFormer | 84.41 | 81.36 | 95.47 | 82.80 | 73.40 |
| CTD-Former | 87.29 | 83.17 | 96.12 | 85.08 | 76.24 |
| CGNet | 85.87 | 85.30 | 96.00 | 85.58 | 76.83 |
| EATDer | 87.16 | 77.18 | 93.84 | 81.16 | 71.18 |
| SAM-CD | 84.42 | 85.38 | 95.78 | 84.89 | 75.93 |
| LightCD-S | 75.91 | 83.22 | 93.38 | 78.97 | 68.74 |
| TinyCD | 75.13 | 76.66 | 93.15 | 75.86 | 65.60 |
| SRA-VFM | **91.44** | **86.52** | **97.09** | **88.91** | **81.21** |

introduces only 1.8 M additional parameters and 1.3 GFLOPs extra computation, yet yields a 1.84% mIoU improvement (90.71% → 92.55%). By comparison, ChangeFormer sacrifices 38 M parameters and 802 GFLOPs for a mere 0.58% gain, while CGNet requires 7.7× more parameters than ours but achieves 0.97% lower accuracy. Thus, the proposed approach, within the family of comparable methods, incurs only modest overhead while delivering the largest accuracy gain, thereby remaining lightweight and highly efficient.

*2) Results on the Levir-CD Dataset:* As evidenced in Table II, our proposed methodology exhibits statistically significant performance superiority over conventional approaches when evaluated on the Levir-CD benchmark dataset. It achieves the best results in five metrics: Pre, Rec, OA, m$F1$, and mIoU. Specifically, our approach outperforms the second-best ChangeCLIP by 0.37% in mIoU. In contrast to SAM-CD, we innovate by adopting a frozen-parameter VFM for feature extraction while incorporating trainable parameters in downstream processing, yielding a 1.84% mIoU boost and 1.92% Rec enhancement.

*3) Results on the WHU-CD Dataset:* Table III presents the comparative results between our proposed method and conventional approaches on the WHU-CD dataset. While achieving the second-best Pre score compared to CGNet, our approach demonstrates superior or equivalent performance across the remaining four evaluation metrics. Notably, our model maintains a compact architecture with less than half the size and fewer than one-seventh the parameters of CGNet.

*4) Results on the CLCD Dataset:* Table IV demonstrates that our proposed approach attains SOTA performance across all five evaluation metrics on the CLCD dataset. Notably, we achieve a 4.38% mIoU improvement over the suboptimal CGNet and a 5.28% enhancement compared to SAM-CD.

*5) Results on the S2Looking Dataset:* Table V reports the benchmarking results of our approach against ten conventional methods on the S2Looking dataset. Across all evaluation metrics, our method consistently achieves either the best or second-best performance. Notably, we surpass the suboptimal BIT method by 3.29% in terms of mIoU.

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TABLE V
RESULTS ON THE S2LOOKING DATASET

| Method | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| FC-Siam-diff | **88.95** | 52.38 | 98.83 | 65.93 | 4.71 |
| FC-Siam-conc | 79.77 | 57.04 | 98.85 | 66.51 | 12.98 |
| SNUNet | 45.26 | 50.60 | - | 47.78 | 31.39 |
| BIT | 70.26 | 56.53 | - | 62.65 | 45.62 |
| ChangeFormer | 72.82 | 56.13 | - | 63.39 | - |
| EATDer | 65.85 | 54.74 | - | 59.78 | 42.64 |
| SAM-CD | 87.96 | 75.89 | 99.23 | 80.76 | 44.82 |
| LightCD-S | 82.68 | 73.98 | 99.07 | 77.64 | 38.65 |
| TinyCD | 86.26 | 75.58 | 99.18 | 79.97 | 43.22 |
| IAE-CDNet | 86.69 | 66.50 | - | - | - |
| SRA-VFM | 88.44 | **78.43** | **99.28** | **83.13** | **48.91** |

TABLE VI
RESULTS ON THE SYSU-CD DATASET

| Method | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| CDNet[50] | - | - | 89.90 | 85.86 | 75.99 |
| SNUNet | - | - | 90.79 | 86.80 | 77.41 |
| L-UNet[51] | - | - | 90.58 | 86.75 | 77.30 |
| IFN[52] | - | - | 90.74 | 87.06 | 77.75 |
| SAM-CD | 86.21 | **87.91** | 90.41 | 87.01 | 77.62 |
| LightCD-S | 80.52 | 81.38 | 86.06 | 80.93 | 69.14 |
| TinyCD | 73.30 | 79.10 | 79.31 | 74.84 | 60.95 |
| FTA-Net[48] | 85.91 | 82.33 | - | - | - |
| SRA-VFM | **89.11** | 87.62 | **91.75** | **88.36** | **79.67** |

*6) Results on the SYSU-CD Dataset:* Table VI shows the comparison of our method with eight conventional methods. In terms of m$F1$ and mIoU, our method outperforms IFN by 1.3% and 1.92%, respectively. In terms of OA and Pre, our method outperforms the suboptimal methods by 0.96% and 2.9%.

### D. Ablation Study

*1) Selection of the Backbone Network:* To verify the encoder's feature extraction capability impact on model performance, we conduct ablation experiments on the LEVIR-CD dataset using three FastSAM variants: FastSAM-x, FastSAM-s, EfficientSAM [53], and Prithvi [25]. By default, our method employs FastSAM-x as the visual encoder. Compared to FastSAM-s, FastSAM-x is a larger model potentially possessing more parameters, enabling it to capture more complex image features. However, correspondingly, FastSAM-x may also require more computational resources and memory. The features extracted from FastSAM-x and FastSAM-s at four stages have the same image dimensions but differ in the number of channels. Specifically, the feature channel counts extracted by FastSAM-x are {160, 320, 640, 640}, while those for FastSAM-s are {64, 128, 256, 512}. EfficientSAM, a lightweight vision foundation segmentation model introduced by Meta, achieves high accuracy while substantially reducing computational complexity and memory demands through masked image pretraining and an optimized lightweight Vision Transformer (ViT) architecture, exhibiting exceptional performance in zero-shot instance segmentation. Prithvi is a geospatial foundation model that leverages a masked autoencoder framework, pretrained on multispectral temporal HLS data over the contiguous United States, and delivers rapid fine-tuning together with data-efficient performance across diverse downstream Earth observation tasks worldwide. As shown in the experimental results in Table VII, FastSAM-x outperforms FastSAM-s on all evaluation metrics, particularly achieving a 0.55% improvement in the mIoU metric. We attribute this performance enhancement to the increased number of learnable parameters in FastSAM-x. Additionally, our method still surpasses other methods when using the more lightweight FastSAM-s, demonstrating the superiority of our approach. Likewise, the experimental results reveal a performance degradation when EfficientSAM is employed as the backbone network, primarily attributable to its parameter count being approximately 20-fold lower than that of FastSAM, thereby diminishing the model's capacity for feature fitting.

TABLE VII
SELECTION OF THE BACKBONE NETWORK

| Method | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| FastSAM-S | 96.44 | 94.97 | 99.18 | 95.70 | 92.00 |
| EfficientSAM | 90.11 | 91.83 | 98.21 | 90.96 | 84.42 |
| Prithvi | 91.34 | 89.16 | 98.16 | 90.22 | 83.36 |
| SRA-VFM | **96.54** | **95.49** | **99.24** | **96.01** | **92.55** |

TABLE VIII
NUMBER OF DATA AUGMENTATION SLICES

| Numbers | Pre | **Rec** | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| 4 | 96.41 | 95.39 | 99.21 | 95.90 | 92.35 |
| 8 | 96.24 | 95.43 | 99.20 | 95.83 | 92.25 |
| 16 | 96.39 | **95.50** | 99.22 | 95.94 | 92.44 |
| 32 | 96.50 | 95.19 | 99.21 | 95.84 | 92.25 |
| {4, 8, 16, 32} | **96.54** | 95.49 | **99.24** | **96.01** | **92.55** |

TABLE IX
DATA AUGMENTATION METHODS

| Method | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| random_crop | 96.64 | 86.45 | 98.42 | 91.26 | 84.30 |
| random_flip | 94.49 | 94.11 | 98.90 | 94.30 | 89.66 |
| cutmix | 96.27 | 94.73 | 99.14 | 95.49 | 91.65 |
| mixup | **97.19** | 88.92 | 98.70 | 92.87 | 86.97 |
| cut-regroup | 96.54 | **95.49** | **99.24** | **96.01** | **92.55** |

*2) Number of Data Augmentation Slices:* To enhance dataset diversification and facilitate change feature recognition in the network, we custom-design a slice-reassemble data augmentation method for RSCD tasks, as illustrated in Fig. 2. In this method, different slice sizes preserve varying levels of fine-grained local image correlations. To systematically evaluate spatial partitioning impacts, we conduct ablation studies on the reassembly module across varying slice granularities. The experimental results in Table VIII reveal that different slice sizes can improve model performance, with the best performance enhancement achieved by randomly combining slices of different sizes. This is because the random selection approach maximizes the diversity of the input images. Consequently, our method adopts a combination of slice sizes of {4, 8, 16, 32}.

*3) Data Augmentation Methods:* In Table IX, we compare the SRA with other commonly used data augmentation techniques in visual tasks. Specifically, we implement four canonical data augmentation strategies for experimental comparison: Random Crop, Random Flip, CutMix, and MixUp

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TABLE X
SELECTION OF LOSS BALANCE FACTOR

| Loss | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| 1 | 96.43 | 94.66 | 99.15 | 95.54 | 91.72 |
| 2 | 96.14 | **95.49** | 99.20 | 95.81 | 92.21 |
| 5 | 96.43 | 95.41 | 99.22 | 95.92 | 92.39 |
| 10 | **96.54** | **95.49** | **99.24** | **96.01** | **92.55** |
| 20 | 96.42 | 95.28 | 99.21 | 95.84 | 92.26 |

TABLE XI
NUMBER OF RESBLOCKS FOR BOUNDARY FEATURE

| Method | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| 2 | 96.37 | 95.49 | 99.22 | 95.93 | 92.41 |
| 4 | **96.54** | 95.49 | **99.24** | **96.01** | **92.55** |
| 6 | 96.03 | **95.80** | 99.21 | 95.91 | 92.38 |
| 8 | 96.44 | 95.16 | 99.21 | 95.85 | 92.26 |

TABLE XII
ABLATION OF DIFFERENT COMPONENTS

| SRA | DFED | MSH | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|---|---|
| - | - | - | 95.08 | 93.74 | 98.93 | 94.40 | 89.82 |
| ✓ | - | - | 95.63 | 94.85 | 99.09 | 95.24 | 91.23 |
| - | ✓ | - | 95.74 | 95.09 | 99.12 | 95.41 | 91.52 |
| - | - | ✓ | 95.57 | **95.65** | 99.15 | 95.61 | 91.86 |
| ✓ | ✓ | - | 96.28 | 95.44 | 99.21 | 95.86 | 92.29 |
| ✓ | ✓ | ✓ | **96.54** | 95.49 | **99.24** | **96.01** | **92.55** |

TABLE XIII
ABLATION OF ENB MODULE

| EnB Module | Pre | Rec | OA | mF1 | mIoU |
|---|---|---|---|---|---|
| w/o | 94.88 | 92.94 | 98.61 | 93.39 | 89.11 |
| w/ | **96.54** | **95.49** | **99.24** | **96.01** | **92.55** |

Random Crop improve model generalization through stochastic spatial cropping during training. Random flip augments the dataset and improves model robustness by randomly flipping the input image. CutMix enriches dataset diversity and boosts model robustness and generalization by combining partial regions of two samples in proportion. MixUp simulates complex interactions between samples by linearly interpolating two or more input samples to generate new training samples. Although these four methods perform well on visual tasks such as classification and object detection, the experimental results show that their performance improvements are relatively limited for RSCD tasks. In contrast, our customized slice-reassemble method achieves better results by preserving the local semantics of the input bi-temporal RSIs while enriching the diversity of the dataset even further.

*4) Selection of Loss Balance Factor:* To better capture the contours of the CD areas and further enhance model performance through multitask joint optimization, our method not only detects and segments the RSI change areas but also predicts the contours. Therefore, our method includes two supervised loss functions: one for CD and one for contour segmentation. To balance these two loss functions and achieve better model performance, we apply a weighting setting to the contour segmentation auxiliary task. In Table X, we conduct experiments on the selection of the hyperparameter $\beta$ for the weight. The experimental results show that different hyperparameters have a significant impact on the model. When $\beta$ is set to 10, the model achieves optimal performance. Given that contour segmentation focuses on pixel-level refinement at boundaries rather than comprehensive CD, we assign a higher weight coefficient to prioritize model attention on these critical regions. This adjustment accounts for the task's inherent requirement for precise edge localization beyond bulk change identification. However, excessive $\beta$ values cause the model to prioritize the auxiliary contour segmentation task entirely, thereby diminishing main task supervision.

*5) Number of Resblocks for Boundary Feature:* In our change region segmentation head network, after merging the $O^2_{D1}$ and $O^2_{D2}$ features, we use residual modules for further feature extraction. The number of residual modules determines the parameter count and depth of this feature extraction network. In Table XI, we conduct comparative experiments using different numbers of residual modules. Empirical analysis reveals that excessive residual module stacking induces performance degradation through overfitting stemming from parameter redundancy. Our experiments identify four residual modules as the optimal configuration, achieving peak performance in CD tasks. Consequently, we establish this configuration as the default architecture in our framework.

*6) Ablation of Different Components:* In the context of RSCD tasks, we propose three innovative components: the SRA, the DFED, and the MSH. Empirical validation through ablation studies (Table XII) systematically analyzes individual module contributions and synergistic effects on model performance enhancement. The experimental results show that using the slice-reassemble augmentation (SRA), top-down feature adapter, and change region segmentation head network modules individually can achieve performance improvements of 1.41%, 1.7%, and 2.04% in mIoU, respectively, compared to the baseline method. When different components are used simultaneously, the model can achieve further performance enhancements. Ultimately, our method outperforms the baseline method on all evaluation metrics, with an improvement of 2.73% relative to the baseline. This experiment fully demonstrates the effectiveness and superiority of our proposed method.

*7) Ablation of EnB Module:* Within our RSCD framework, we design the EnB module to adaptively recalibrate low-level features extracted by FastSAM, ensuring optimal alignment with remote sensing CD requirements. To assess the module's capacity in bridging semantic gaps between general visual tasks and domain-specific CD, we conduct an ablation study detailed in Table XIII. This experiment compares the EnB architecture against baseline downsampling operations and $1 \times 1$ convolutions. Empirical results confirm that integrating the EnB module significantly enhances model performance, primarily attributable to its innovative dual-branch design. This architecture synergizes CNN's detail-preserving capabilities for fine-grained feature representation with Mamba's global context modeling for long-range dependency capture. Such integration substantially improves the framework's adaptability to complex remote sensing scenarios characterized by heterogeneous spatial patterns and multiscale variations.

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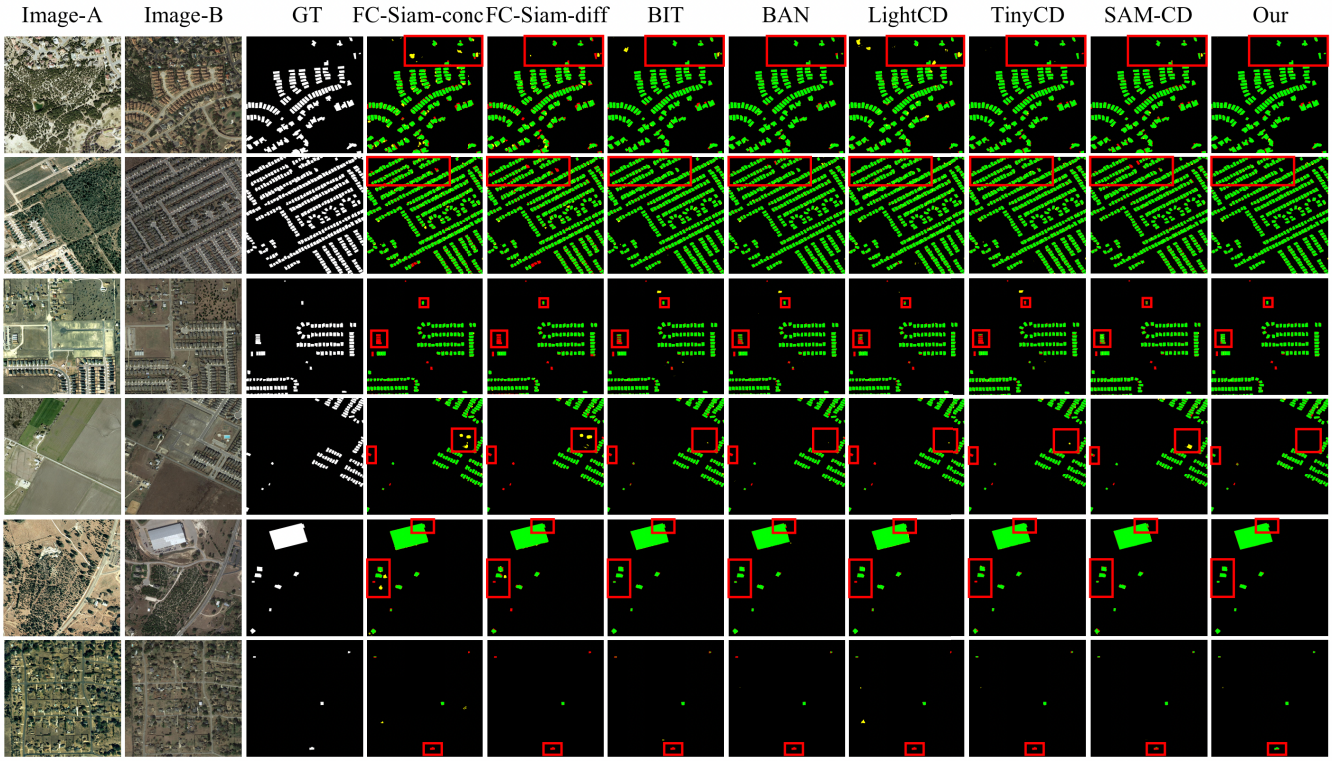



Fig. 5. Visualization results of our method on the LEVIR-CD dataset.

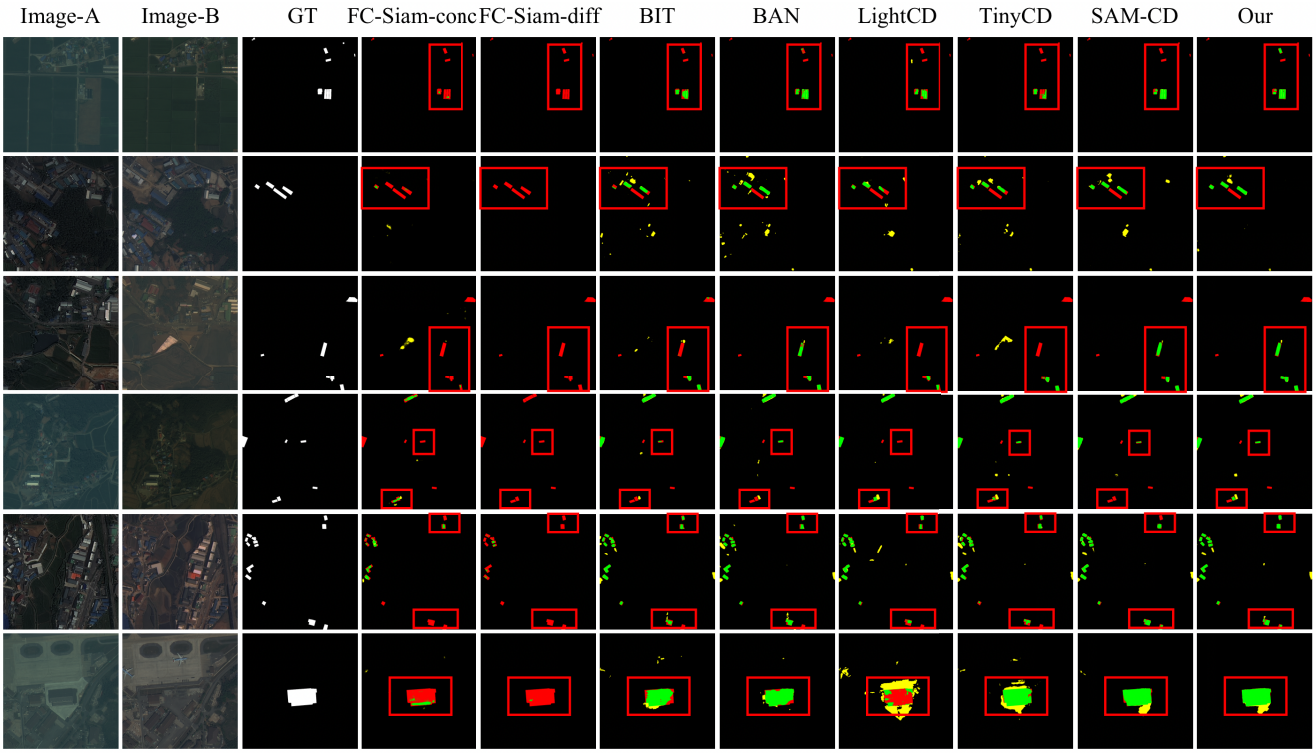



Fig. 6. Visualization results of our method on the S2Looking dataset.

### *E. Visualization Presentation*

Fig. 5 shows the experimental visual results on the LEVIR-CD dataset. From the first row to the fourth row, it is evident that our method has the fewest FPs and false negatives when detecting changes in dense buildings. The fifth row shows that in images containing both large-scale and small-scale buildings, our method, along with BIT, BAN, TinyCD, and SAM-CD, detects building changes. However, our method

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provides the most accurate building edges due to the integrated change region contour edge detection auxiliary task. The sixth row shows that our method also achieves the best results in detecting small-scale buildings.

Fig. 6 shows the experimental visual results of our method compared to other common methods on the S2Looking dataset. Compared to other methods, it is clearly evident that our method detects more accurate building changes. From the first five rows, it can be seen that other methods, influenced by factors such as lighting and bare land, often misclassify unrelated areas as changed buildings. Our design makes the visual foundation model more suitable for RSI scenarios, effectively reducing these misclassifications. The last row shows that for large building changes, common methods struggle to accurately delineate the regions of change. Thanks to our designed edge detection auxiliary task, our method can better detect the edge information of large changed buildings.

## IV. Conclusion

This study introduces a collaboratively optimized framework for RSCD that integrates data augmentation and model adaptation strategies, with a clear emphasis on its broad applicability across diverse remote sensing scenarios. By leveraging slice-reassemble data augmentation and hierarchical feature adaptation mechanisms, we substantially enhance the accuracy and robustness of bi-temporal RSCD systems, providing a versatile tool for urban expansion monitoring, ecological environment assessment, and other land cover dynamics analyses. Experimental validation demonstrates that our approach surpasses SOTA methods across multiple benchmark datasets, underscoring its generalizability and practical value. While the pretrained model-based feature alignment mechanism partially mitigates domain discrepancies, cross-domain generalization for multimodal remote sensing data [e.g., synthetic aperture radar (SAR) and optical imagery] requires rigorous validation. Future research will focus on developing cross-modal feature decoupling networks and alignment architectures to establish a unified CD framework capable of processing multisource data, including optical and SAR images. Overall, this work establishes a novel technical paradigm for intelligent remote sensing interpretation, with demonstrated scalable applicability to diverse scenarios, including urban expansion monitoring and ecological environment assessment.

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